

Louis James Vasseur

MSc Computer Science, AI & Data Science, EPFL

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EXPERIENCE & EDUCATION

BSc Computer Science - EPFL

Sept 2022 – Jul 2025

MSc Computer Science - EPFL · AI & Data Science specialization · Current GPA 5.67/6

Sept 2025 – now

Research Intern - EPFL LIAC Lab (continuation of STRUCTURE Project work)

Jul – Aug 2026

- Structure elucidation: identifying an unknown molecule from its spectra (mass spectrometry, infrared, $^1\text{H}/^{13}\text{C}$ NMR, HSQC), a daily bottleneck in chemistry labs
- Built and compared multiple decoder families in PyTorch (autoregressive sequence and discrete graph diffusion) against a shared multimodal spectral encoder.
- Designed the evaluation and ablation protocol now used to rank decoders (eg. per-modality attribution, etc...)

Research Lead + Project Lead - EPFL AI Team

Aug 2025 – now

- Run the research track of a student-led AI lab: 10+ projects, 50+ members across neuroscience, chemistry, sign language, spectroscopy, earth science, agentic AI.
- Set technical direction and review across teams and opened collaborations with EPFL labs (NeuroAI, LIAC), CERN and CHUV.
- As Project Lead, took GESTURE (team of 9) to a live demo at AMLD 2026 and currently leading BRAIN (team of 6).

LLM Benchmarking - Numenta

Aug 2024

- Quantify what LLM inference costs on different operating-system architectures: latency and throughput benchmarking across 3 OS architectures for summarization and retrieval-augmented generation, with cross-platform reports and comparative visualizations.

PUBLICATIONS

step2point: Optimising point-cloud representations for fast calorimeter simulation

A. Zaborowska, P. McKeown, P. Raikwar, M. Mozzanica, L. J. Vasseur, M. Casadei, M. Casiraghi, S. Diallo, S. Chopra, T. Tej Nandi

CHEP 2026 - Bangkok - Oral presentation - May 2026

TAG-X: Joint Multispectral Representations for Molecular Structure Elucidation

L. J. Vasseur, R. Gautam, M. Gfeller, J. Barozet-Golbery, R. Schlama, P. Schwaller

NeurIPS 2026 Workshop on Agentic Systems for Molecular Sciences (ML4Molecules) - Paris (accepted)

PROJECTS

Compressed Learned Indexes at Scale (working with EPFL DIAS Lab)

Sept 2026 – now

- Developing a C++20 compressed learned index jointly fitting gap compression, a monotone normalizing-flow key transform and virtual points on a single least-squares loss.
- Evaluating joint vs. individual (NFL, CSV) and sequential transforms in lookup comparisons, memory and build time: 2M-key samples on 10 datasets; 200M-key GRE runs against ALEX, LIPP, PGM, ART and B+tree in preparation.

Project WORLD - evaluating NVIDIA surgical video world models

Sept 2026 – now

- Evaluating Cosmos-H-Surgical-Simulator against real Open-H-Embodiment robotic surgery, pairing rollouts with the same initial frame and robot kinematics to assess tissue response and, later, instrument motion.
- Developing a Waymo Sim Agents likelihood-based metric over tracked tissue motion: camera-compensated velocity, strain energy and planned instrument-contact response; KLT with CoTracker3 cross-checks, validation ongoing.

Project STRUCTURE - reading molecules out of spectra (working with EPFL LIAC Lab)

Feb 2026 – now

- Chemistry foundation model aiming to map a molecule's measured spectra directly to its chemical structure, replacing manual expert interpretation.
- Set-transformer encoder over irregular spectral peaks, contrastively aligned to a chemical language model (MoLFormer-XL) with multiple self-supervised objective, multiple decoder families compared (eg autoregressive, graph diffusion, flow matching...) over 715k molecules.
- 80%+ top-1 and 90%+ top-10 exact-structure recovery on held-out molecules

Project BRAIN - decoding what people see and hear from EEG (working with EPFL NeuroAI Lab)

Feb 2026 – now

- NeuroAI: reconstruct the image or music a person is perceiving from non-invasive brain recordings.
- Transformer encoders over EEG time series aligned to CLIP/CLAP embedding spaces by contrastive learning, LoRA-adapted foundation-model features, and diffusion image generation for reconstruction.
- Shipped a live demo running end-to-end on a physical EEG system.

Project GESTURE - real-time sign-language translation

Feb 2025 – Feb 2026

- Accessibility: two-way, real-time translation between American Sign Language and text.
- Graph convolutional network (ST-GCN) over MediaPipe pose skeletons for recognition; diffusion motion generation with anatomical priors (SMPL-X, VPoser), knowledge distillation and LLM beam search (Qwen3) for word reordering.
- Distillation halved generation loss (0.64 $\rightarrow$ 0.31); deployed as a live Flask + Socket.IO + React demo with live avatar, presented at AMLD 2026 @ EPFL.

Dark Matter Density Generation - EPFL Lab of Astrophysics

Oct – Dec 2025

- Astrophysics: produce realistic galaxy-cluster mass maps from physical parameters alone, without running a simulation.
- Contrastive + linear-probe dual objective, foundation-model finetuning, flow matching and a Diffusion Transformer on image-like scientific data.
- Map generation cut from potential days of numerical simulation to minutes/hours of inference

BSc Thesis - early cancer warning signs

Feb – Jun 2025

- Clinical ML: flag early cancer warning signs from patient profiles in a small, sparse medical dataset.
- Custom gradient-boosted tree ensemble with feature engineering built for sparse tabular clinical data.

TECHNOLOGIES

Languages Python, C, C++20, Java, Scala, SQL, JavaScript

ML / DL PyTorch, TensorFlow, Hugging Face, Weights & Biases, XGBoost, CatBoost, RDKit, MNE-Python, MediaPipe

Methods Transformers, graph neural nets, diffusion and flow matching, contrastive and self-supervised learning, LLM finetuning, gradient boosting, retrieval

Data Spectra, EEG, pose and video, scientific images, molecular graphs, sparse clinical tables

Tooling SLURM (multi-GPU HPC), Docker, Git, Linux, Flask, Socket.IO, React, Pandas, NumPy, D3.js